Nour Taha

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Education

Badr University in Cairo, Cairo, Egypt
B.Sc. in Mechatronics Engineering 2023–2028
GPA: 3.48/4.0
Relevant Coursework: Data Acquisition (DAQ), Electronics, C++/Python Programming, Statistics, Modeling and Simulation, Machine Design, Control Systems

Projects

Smart Vent — Mechanical Ventilator Prototype

- Designed and implemented a sensor-driven mechanical ventilator system with real-time motor synchronization using C++ and DAQ sensors.
- Processed real-time sensor data to dynamically adjust ventilation parameters for optimal performance.
- Integrated a buzzer alert system for safety monitoring and fault detection.

F1 Lap Time Prediction Model

- Built a machine learning model to predict lap times using motorsport dynamics and data analysis.
- Performed feature engineering, preprocessing, and model training using Scikit-learn and Pandas.
- Optimized model performance through hyperparameter tuning, improving prediction accuracy.

Real-Time Air Quality Monitoring System — LabVIEW & DAQ

- Developed a real-time monitoring system with air quality sensors and DAQ integration.
- Built LabVIEW block diagrams for signal processing and interactive dashboards.
- Implemented sensor interfacing and data acquisition for reliable environmental monitoring.

Robotics Competition — Programming Lead

- Led software development and algorithm implementation for a robotics competition team.
- Developed real-time control algorithms using C++ and embedded systems, enabling the team to secure 1st place in the RC Car Fighting Competition.

CubeSat Satellite — Project Member

- Contributed to the design, development, and integration of a CubeSat satellite as part of a collaborative aerospace project.
- Assisted in testing satellite subsystems and performing validation for mission readiness.

Experience

EGSA — Intern, Cairo, Egypt Aug 2024

- Assisted in the design, development, and testing of satellite subsystems (OBC, ADCS, EPS).
- Conducted system validation tests and collaborated on aerospace applications.
- Gained hands-on experience in control systems, embedded electronics, and data acquisition.

FUZETEK — Intern, Cairo, Egypt Feb 2025

- Applied data structures and algorithms to improve software efficiency and reliability.
- Assisted in debugging, testing, and optimizing embedded software components.
- Developed technical documentation and collaborated with engineers on system integration.

Leadership & Activities

IEEE — Vice Technical Director

- Instructor for the Data Science track, teaching foundational and intermediate Python, data analysis, and ML concepts to volunteers.
- Mentor team members, guiding project development, technical skills, and career growth.
- Lead technical workshops and code reviews to improve volunteer competency.

GDSC — IoT Volunteer

- Recognized as Golden Volunteer for outstanding contributions to IoT projects and community initiatives.
- Designed and developed IoT systems using Arduino and ESP32 with Wi-Fi integration.
- Applied embedded programming and sensor interfacing in real-world IoT solutions.

VEP — Entrepreneur Volunteer

- Designed and developed a comprehensive business plan and Lean Canvas model for GreenGrow, a gardening-focused entrepreneurship project under the VEP program, including market analysis, value proposition, and financial projections

Technical Skills

- **Programming Languages:** Python, C++, MATLAB, SQL
- **Machine Learning & AI:** Scikit-learn, TensorFlow, Pandas, NumPy, OpenCV
- **CAD & Simulation:** SolidWorks, AutoCAD, Simulink
- **IoT & Embedded Systems:** Arduino, ESP32, Sensor Interfacing, DAQ, Real-Time Control
- **Software & Tools:** LabVIEW, PyCharm, Jupyter Notebook, Visual Studio, Git
- **Leadership & Competencies:** Project Management, Team Leadership, Problem Solving, Adaptability